

Linear Regression on California Housing Dataset

Submitted By: Janvi

College Name: Gandhi Memorial National College

Internship Name: AI & ML Internship

Task: Task 1 – Linear Regression Model

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1. Objective

The objective of this project is to build a Linear Regression model using the California Housing Dataset. The model is used to predict house prices based on different housing features. The performance of the model is evaluated using MAE, RMSE, and R² Score.

2. Dataset

The California Housing Dataset is a popular dataset used for machine learning. It contains information about houses in different areas of California. The dataset has 8 input features such as median income, house age, average rooms, average bedrooms, population, average occupancy, latitude, and longitude. The target variable is the median house value (MedHouseVal).

3. Exploratory Data Analysis (EDA)

The dataset was analyzed before training the model. The data contains no missing values. Statistical information was obtained using the describe() function. Correlation between different features was also checked to understand the relationship between variables.

Exploratory Data Analysis (EDA)

- * The dataset was loaded successfully.
- * Dataset information was checked using df.info().
- * Summary statistics were generated using df.describe().
- * Missing values were checked using df.isnull().sum().
- * Correlation between features was analyzed.

df.info()

```
<class 'pandas.DataFrame'>
RangeIndex: 20640 entries, 0 to 20639
Data columns (total 9 columns):
#   Column      Non-Null Count  Dtype
---  -
0   MedInc      20640 non-null  float64
1   HouseAge    20640 non-null  float64
2   AveRooms    20640 non-null  float64
3   AveBedrms   20640 non-null  float64
4   Population  20640 non-null  float64
5   AveOccup    20640 non-null  float64
6   Latitude    20640 non-null  float64
7   Longitude   20640 non-null  float64
8   MedHouseVal 20640 non-null  float64
dtypes: float64(9)
memory usage: 1.4 MB
```

df.describe()

MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude	MedHouseVal	
MedInc	1.000000	-0.119034	0.326895	-0.062040	0.004834	0.018766	-0.079809	-0.015176	0.688075
HouseAge	-0.119034	1.000000	-0.153277	-0.077747	-0.296244	0.013191	0.011173	-0.108197	0.105623
AveRooms	0.326895	-0.153277	1.000000	0.847621	-0.072213	-0.004852	0.106389	-0.027540	0.151948
AveBedrms	-0.062040	-0.077747	0.847621	1.000000	-0.066197	-0.006181	0.069721	0.013344	-0.046701

Population	0.004834	-0.296244	-0.072213	-0.066197	1.000000	0.069863	-0.108785	0.099773	-0.024650
AveOccup	0.018766	0.013191	-0.004852	-0.006181	0.069863	1.000000	0.002366	0.002476	-0.023737
Latitude	-0.079809	0.011173	0.106389	0.069721	-0.108785	0.002366	1.000000	-0.924664	-0.144160
Longitude	-0.015176	-0.108197	-0.027540	0.013344	0.099773	0.002476	-0.924664	1.000000	-0.045967
MedHouseVal	0.688075	0.105623	0.151948	-0.046701	-0.024650	-0.023737	-0.144160	-0.045967	1.000000

4. Model Used

The Linear Regression algorithm was used for this project. The dataset was divided into 80% training data and 20% testing data using the train_test_split() function. The model was trained using the training data and then used to predict house prices on the test data.

5. Results

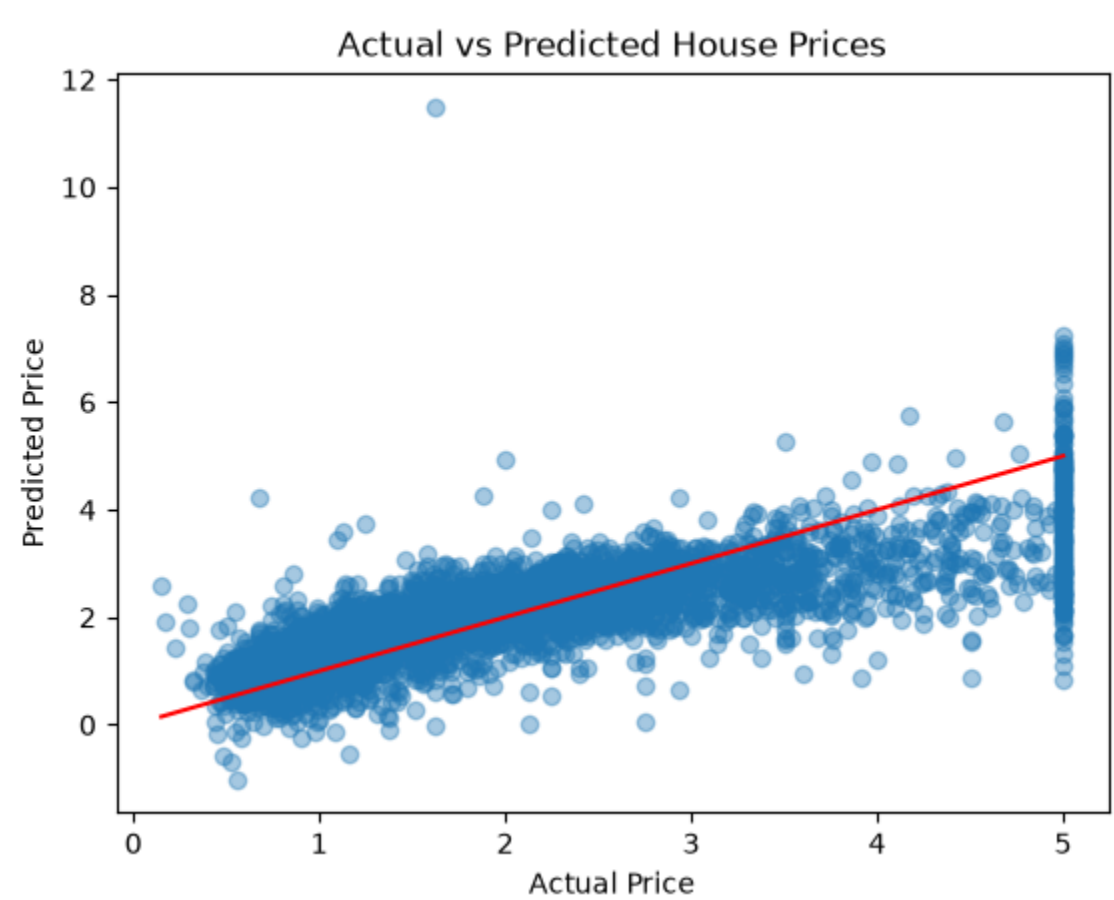
The model performance was evaluated using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and R² Score. An Actual vs Predicted scatter plot was also generated to visualize the prediction performance.

(Add your MAE, RMSE, and R² Score values here.)

MAE: 0.5332001304956555

RMSE: 0.7455813830127763

R2 Score: 0.575787706032451



6. Conclusion

The Linear Regression model was successfully trained on the California Housing Dataset. The model predicted house prices with good accuracy. This project helped in understanding data analysis, model training, prediction, and performance evaluation using machine learnig.

7. Future Improvements

- * Use advanced machine learning models such as Random Forest or XGBoost for better prediction accuracy.
- * Perform feature engineering to create more useful features.
- * Tune the model using hyperparameter optimization.
- * Handle outliers and normalize the data for improved performance.
- * Deploy the model as a simple web application using Flask or Streamlit so users can predict house prices easily.

8. References

1. Scikit-learn Documentation: <https://scikit-learn.org/stable/>

2. California Housing Dataset: https://scikit-learn.org/stable/modules/generated/sklearn.datasets.fetch_california_housing.html

3. NumPy Documentation: <https://numpy.org/doc/>

4. Pandas Documentation: <https://pandas.pydata.org/docs/>

5. Matplotlib Documentation: <https://matplotlib.org/stable/>

6. Seaborn Documentation: <https://seaborn.pydata.org/>